

1.8 Methods for Improving HCI

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CS XII

Chapter 1: Computer Systems (HCI)

Lecture 11 of 15

1.8 Methods for Improving HCI

LEARN IT

Once you can name an HCI problem, you fix it with a matching method. The source lists eight methods. Learn each with its tiny example — they pair almost one-to-one with Section 1.7.

- **Enhance usability:** simplify navigation (a clear "Buy Now" button).
- **Strengthen accessibility:** support disabilities (captions on videos).
- **Ensure consistency:** keep layouts uniform (the same "Save" button everywhere).
- **Provide clear feedback:** show action results ("Payment Successful" popup).
- **Improve adaptability:** responsive design (the website scales to mobile screens).
- **Apply user-centred design (UCD):** focus on real user goals and pilot-test.
- **Use multimodal interaction:** allow touch, voice or gesture inputs combined.
- **Continuous testing:** regularly update based on user feedback.

User-centred design (UCD) means the real user's goals drive every decision, and you trial (pilot-test) the design with those users instead of guessing in an office.

Responsive design means the layout rearranges itself for phone, tablet and PC, instead of staying a laptop-only page.

WORKED EXAMPLE

Worked example: match each JazzCash / portal problem to a fix.

Problem (1.7)	Method (1.8)
Cannot find "Send Money"	Enhance usability — one clear button
No captions on help video	Strengthen accessibility
Search icon changes from page to page	Ensure consistency
Submit freezes with no message	Clear feedback — "Payment Successful"
Portal crashes on mobile	Improve adaptability — responsive layout
Built features nobody asked for	UCD — interview, then pilot-test
Elderly user cannot type a long number	Multimodal — voice + touch
Complaints ignored after launch	Continuous testing and updates

REMEMBER IT

Think "problem → method" as a pair. Complexity ↔ usability. No captions ↔ accessibility. Different icons ↔ consistency. Freeze ↔ feedback. Phone crash ↔ adaptability. Guessing ↔ UCD. One input only ↔ multimodal. Launch and forget ↔ continuous testing.

COMMON MISTAKE

Mistake: Naming methods without saying *what they change* ("we will improve HCI" is not an answer).

Fix: Always attach the source example: "Provide clear feedback, for example a Payment Successful popup."

اردو میں سمجھیں

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CHECK YOURSELF

1. What does UCD stand for, and what extra step does it include?
2. Which method stops a website from failing on a phone?
3. Give the source's example of clear feedback.

Answers: 1. User-Centred Design; focus on real user goals and pilot-test. 2. Improve adaptability (responsive design). 3. A "Payment Successful" popup.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Improving HCI often starts with:

- a. Watching real users (usability test)
- b. Adding more clipart
- c. Reducing contrast
- d. Hiding labels

MCQ 2. Consistency means:

- a. Same words/icons/positions for the same action
- b. A new metaphor every page
- c. Random colours
- d. Surprise menus

MCQ 3. Simple IA (information architecture) means:

- a. Fewer, clearer paths to tasks
- b. Deeper 12-level menus
- c. All links in a PDF dump
- d. No search

MCQ 4. Prototypes help you:

- a. Fix HCI cheaply before code
- b. Skip users
- c. Skip accessibility
- d. Skip feedback

MCQ 5. Training is a fix when:

- a. The task is rare/complex — but do not use it to excuse a bad UI
- b. The button is invisible
- c. Contrast is 1:1
- d. Errors are in hex only

MCQ 6. Iterative improvement is:

- a. Test → change → test again
- b. Ship once never look
- c. Waterfall with no users
- d. Only career slides

Extra short questions

1. Five methods to improve HCI.
2. Why test with real students not only the developer?
3. A quick win for a form.
4. What is a heuristic evaluation?

Extra long question

1. Take a 7-step fee-payment UI. Cut it to 3 steps, list the HCI problems you removed, and how you would re-test.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Simplify, consistent patterns, better feedback, accessibility, user tests, prototypes, reduce steps.
2. Developers know the map; users get lost.
3. Show required fields, inline errors, remember the class dropdown.
4. An expert checks the UI against rules (consistency, feedback...) before/besides user tests.

Long

1. Steps: amount → confirm → OTP. Removed: extra account-type screens, jargon, no progress. Re-test: 5 students timed, errors counted, one interview.